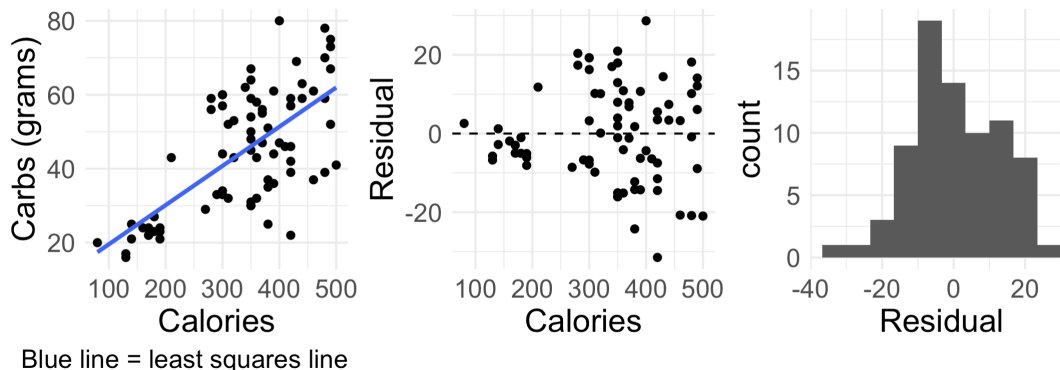


This problem set covers material from Week 10, dates 11/10 – 11/13.

Instructions: Write or type complete solutions to the following problems and submit answers to the corresponding Canvas assignment. Your solutions should be neatly-written, show all work and computations, include figures or graphs where appropriate, and include some written explanation of your method or process (enough that I can understand your reasoning without having to guess or make assumptions). A general rubric for homework problems appears on the final page of this assignment.

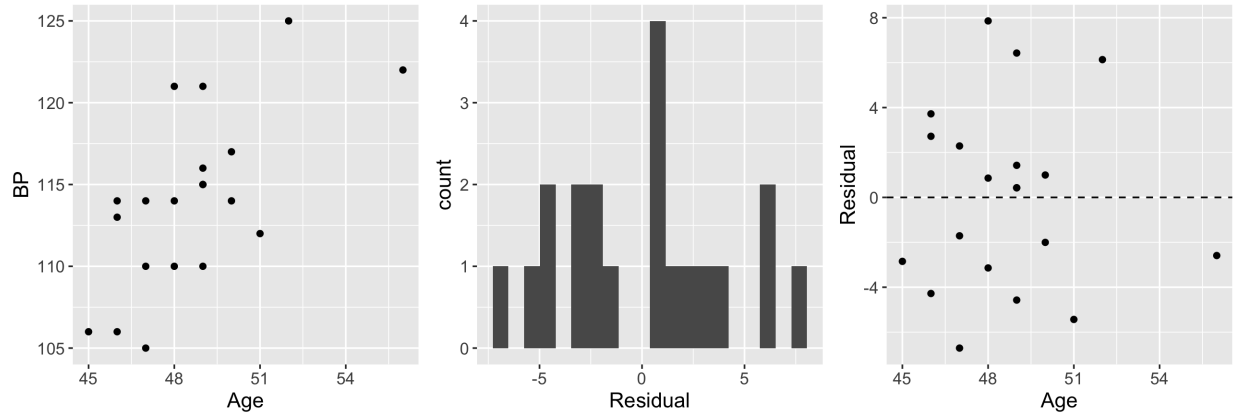
Monday 11/10

1. We will re-visit the `starbucks` data from `openintro`. Since Starbucks only lists the number of calories on the display items in stores, we may be interested in predicting the amount of carbs a menu item has based on its calorie content. The scatterplot below shows the relationship between the number of calories and amount of carbohydrates (in grams) Starbucks food menu items contain.

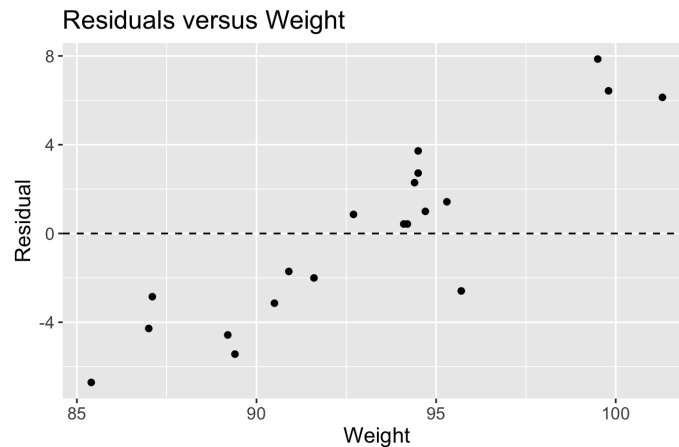


- (a) Describe the relationship (strength, direction, linearity) between number of calories and amount of carbohydrates (in grams) that Starbucks food menu items contain.
 - (b) The least-squares line shown in the first plot is obtained from estimates $b_0 = 8.944$ and $b_1 = 0.106$. Write out two equations *in context*: 1) the linear regression for these data and 2) the fitted model for these data.
 - (c) The menu item Morning Bun has 350 calories, 16 grams of fat, and 45 grams of carbohydrates. Based on your model in (c), obtain the residual for the Morning Bun and explain the meaning of this residual value in context.
 - (d) Do these data meet the conditions required for fitting a least squares line? Check if each condition is met or not, providing brief justification where appropriate.
2. A researcher is interested in determining if a person's age and/or weight are good predictors of their diastolic blood pressure (BP) (specifically for individuals with high blood pressure). They have data on 20 randomly sampled men with high BP, and

for each individual they have recorded the BP, Age, and Weight. The researchers fit a simple linear regression model for BP using Age as the explanatory variable. A scatterplot of the data is shown below, as are the histogram of the residuals and the residual plot.



- Do these data meet the conditions required for fitting a least squares line? Check if each condition is met or not, providing brief justification where appropriate.
- Next, the researchers would like to consider the Weight of each individual as a predictor for BP. They take the residuals from the model above (where Age was the explanatory), and plot those residuals against the individuals' Weight values (shown below). Interpret what this plot is telling you, in context of the data and the fitted model.



- Based on your response in (b), do you think it would be worth considering a model that *also* includes Weight as an explanatory variable, in addition to Age (we don't know how to do that yet). Why or why not?

Wednesday 11/12

TBD

Thursday 11/13

TBD

General rubric

Points	Criteria
5	The solution is correct <i>and</i> well-written. The author leaves no doubt as to why the solution is valid.
4.5	The solution is well-written, and is correct except for some minor arithmetic or calculation mistake.
4	The solution is technically correct, but author has omitted some key justification for why the solution is valid. Alternatively, the solution is well-written, but is missing a small, but essential component.
3	The solution is well-written, but either overlooks a significant component of the problem or makes a significant mistake. Alternatively, in a multi-part problem, a majority of the solutions are correct and well-written, but one part is missing or is significantly incorrect.
2	The solution is either correct but not adequately written, or it is adequately written but overlooks a significant component of the problem or makes a significant mistake.
1	The solution is rudimentary, but contains some relevant ideas. Alternatively, the solution briefly indicates the correct answer, but provides no further justification.
0	Either the solution is missing entirely, or the author makes no non-trivial progress toward a solution (i.e. just writes the statement of the problem and/or restates given information).
Notes:	For problems with multiple parts, the score represents a holistic review of the entire problem. Additionally, half-points may be used if the solution falls between two point values above.
Notes:	For problems with code, well-written means only having lines of code that are necessary to solving the problem, as well as presenting the solution for the reader to easily see. It might also be worth adding comments to your code.